ACWAPOWER · MLPROJECT

Predicting Failures in Green Hydrogen Electrolyzers

Using Machine Learning to Improve Efficiency & Reduce Operational Costs

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THE CHALLEN GE

GreenHydrogen&ACWA Power

Green hydrogen is a clean energy source and a key component in the transition to sustainability and achieving net zero carbon neutrality.

9,985	Sensor Readings
20	Sensor Features
1	Electrolyzer Stack

Problem Statement

Frequent Breakdowns

Electrolyzers suffer frequent shutdowns due to fluctuating operating conditions — increasing maintenance costs.

Efficiency Loss

Each unexpected failure reduces hydrogen output and strains operational budgets at ACWA Power.

ML Solution

Can machine learning predict failures before they occur — enabling proactive maintenance and improved uptime?

Project Goal

Build a classification model that flags risk levels from real-time sensor d ata .

DATASET & PREPROCESSING

01 Raw Excel File

Original data delivered as Excel (.xlsx) from ACWA Power's electrolyzer sensors

02 Convert to CSV

Reformatted to .csv for compatibility with Python ML pipeline & Pandas

03 Column Handling

Renamed columns, fixed data types, removed irrelevant/empty fields

04 Removed ~2,000 Duplicates

Identified and dropped ~2,000 duplicate rows from original ~12,000 records

05 Unlabeled Data

Dataset had no target labels — required a custom labeling strategy before modeling

Final Dataset: **9,985 rows · 20 features · No pre-existing labels**

TWO LABELING APPROACHES

APPROACH 1 — Clustering (K-Means)

- Applied PCA → reduced to 3 components (96.8% variance)
- K-Means with k=4: Silhouette = 0.72, DB = 0.55
- 4 risk classes: Normal, Degradation Warning, Gas Contamination Risk, Critical Failure
- Used domain knowledge to map clusters to severity
- SMOTE applied to balance the minority class (5%)
- Best accuracy: 98.9% (Gradient Boosting + 5-fold CV)
- ! Some class confusion in the classification report

Not selected — class mixing

VS

APPROACH 2 — Threshold-Based √ Chosen

- Applied engineering safety thresholds from domain knowledge
- O_2 in H_2 > 1%, H_2 in O_2 > 1.5%, LDI > 1.5 → label = 1
- Binary labels: Maintenance Needed (44%) / Not Needed (56%)
- Cleaner, physically meaningful label definitions
- Models tested with/without SMOTE and cross-validation
- Best CV accuracy: 86.6% — but very high interpretability
- √ Clear, stable, non-ambiguous class boundaries

√ SELECTED — Best interpretability & reliable boundaries

EXPLORATORY and DATA ANALYSIS

Electrical Features

- Current: 0-8,001 A | Voltage: 0-640 V
- Strong spike at zero = idle periods
- Active ops cluster near 7,800-8,000 A (near full load)
- Current, Voltage & Power: r > 0.95 (physically linked)

Thermal Features

- Room temp stable at 24-25.75 C (controlled env.)
- H2 & O2 outlet temps rise to 80-90 C when active
- Lye supply temp lags outlet temps slightly
- All thermal features drop to room temp when idle

Lye & Separator

- Lye concentration: 0 idle / 28-32 wt% active (normal AWE)
- LDI > 1.5 in some readings = diaphragm pressure risk
- H2 & O2 separator levels move almost identically
- LDI is the most critical safety indicator in this group

Gas Purity & Safety

- O2 in H2 safety limit: 0.5% | H2 in O2: 1.5%
- Some readings exceed limits = diaphragm degradation
- O2 separator pressure clusters at 15-16 bar (active)
- These features drive the threshold labeling st rate g y

Flow & Water Quality

- H2 flowrate & DM water flow: bimodal (0 idle, high active)
- DM water conductivity should stay < 1 uS/cm
- Elevated conductivity = electrolyte contamination risk
- Conductivity is a key early-warning indicator

Correlation Heatmap

- Electrical features strongly correlate with thermal features
- H2 flowrate & DM water flow correlate with current
- Gas crossover features show weak correlation = independent signal
- Idle rows drive a large cluster near-zero across all features

Key insight: Idle vs. Active state is the dominant split in the data — all features show a clear bimodal distribution

MODELLING & RE SU LT S

Pipeline: Threshold Labels → Train/Test Split (80/20) → 5-fold CV → Evaluate

86.6% — **Best Accuracy**
Logistic Regression

0.866 — **Best F1 Score**
Weighted average across classes

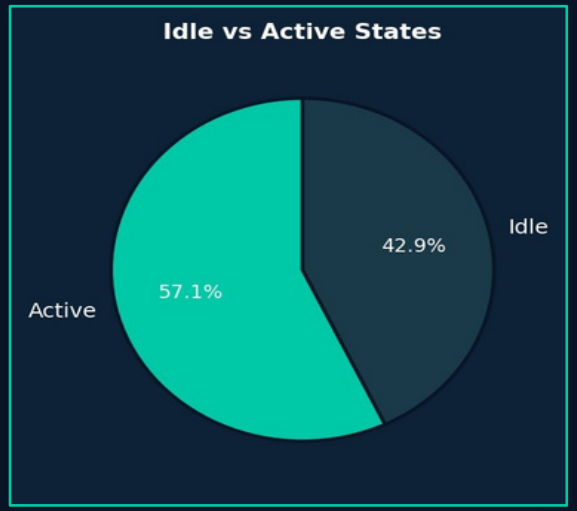
80 / 20 — **Train / Test Split**
Stratified, reproducible

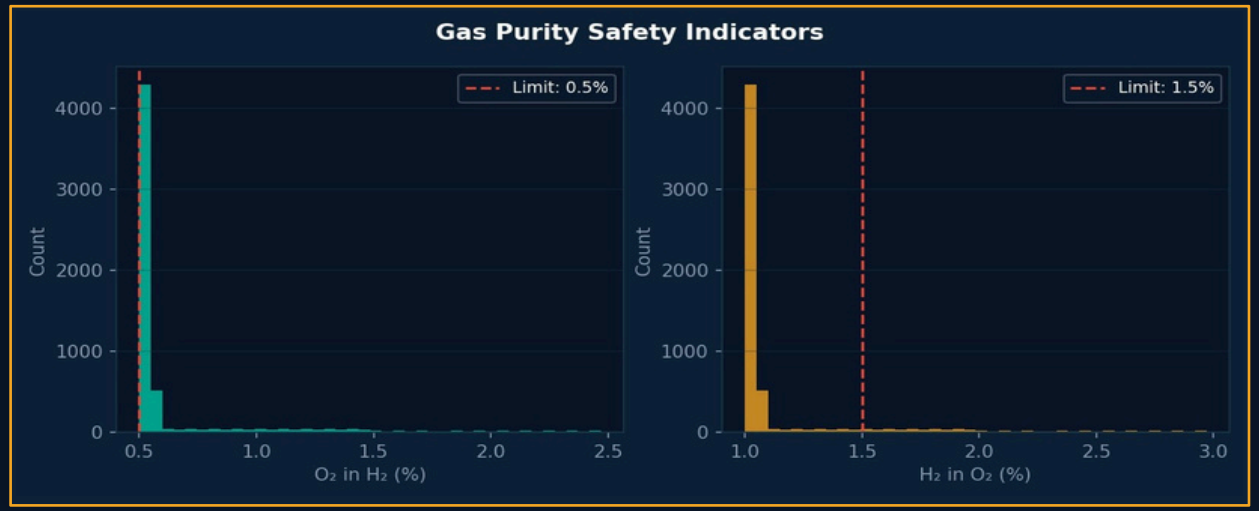
5 — **CV Folds**
StratifiedKFold cross-validation

Models Used:
GaussianNB · BernoulliNB · Logistic Reg. · SVC · KNN · Random Forest · Gradient Boosting · ExtraTrees · AdaBoost · BaggingClassifier

Threshold approach chosen: cleaner labels, stable class boundaries, physically meaningful

EXPLORATORY DATA ANALYSIS

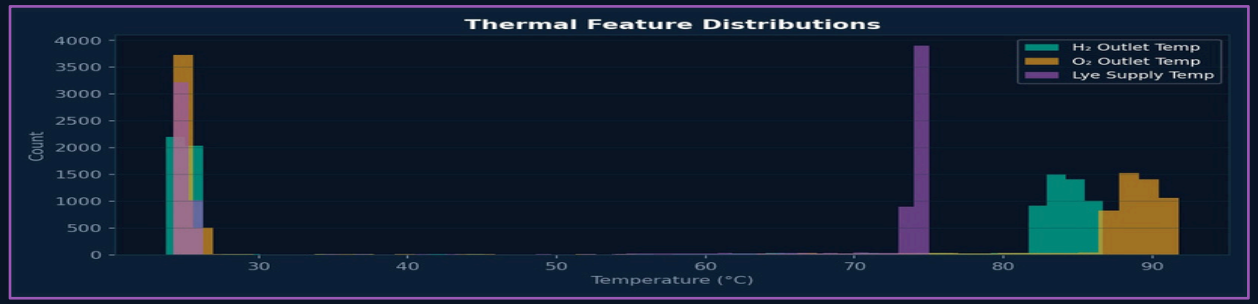


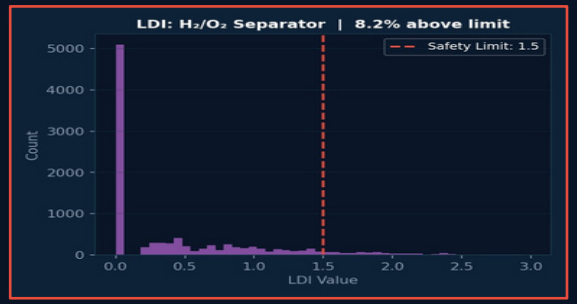


57.6% Active | 42.4% Idle — zeros excluded

Some readings exceed safety limits → triggers label=1

Gas crossover = key maintenance signal





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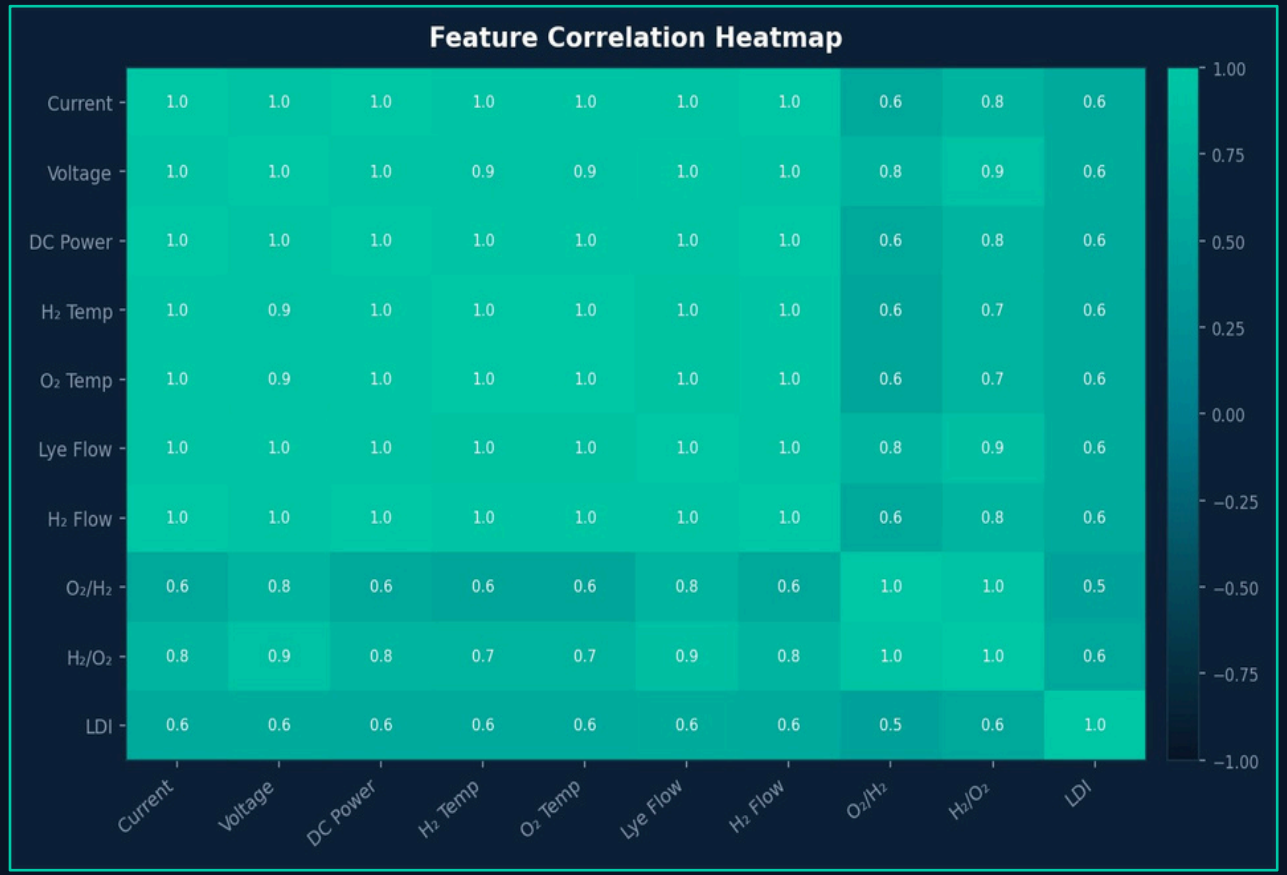

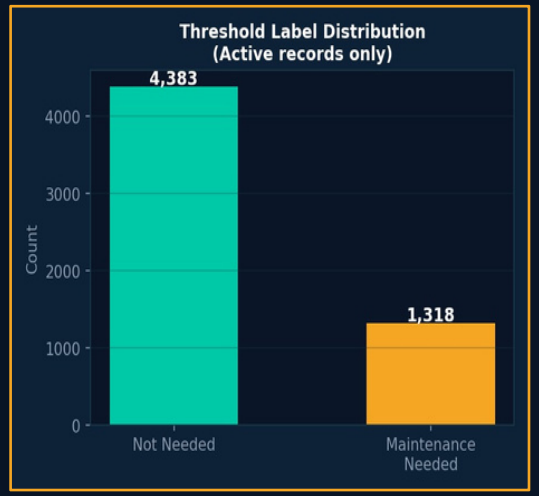

Key Findings

Current / Voltage / Power: r ≥ 0.95

Thermal features follow electrical load

Gas crossover: weak but critical signal

LDI is an independent risk indicator

44% active rows flagged for maintenance

KEYTAKEAWAYS

Conclusion

- ✓ Green hydrogen electrolyzers can be monitored with 20 sensor features to predict maintenance needs before failure occurs.
- ✓ Threshold-based labeling using engineering safety limits produced cleaner, more reliable labels than unsupervised clustering.
- ✓ Best model achieved 86.6% accuracy — sufficient for real-world early-warning systems.
- → Future work: incorporate time-series modeling (LSTM), real-time dashboard, and multi-stack data.